

Steel Beam Estimator

Version 2 - Pipeline Workflow Flowchart

This document summarizes all pipeline phases completed to date, from DXF ingestion through engineering dataset finalization. Phase D.2 (annotation parsing) is the planned next step.

Outputs live under `data/output/` grouped by phase subdirectory. Canonical paths: `src/config/output_paths.py`

Legend

	Input / source data
	Foundation phases (A-C.5)
	Annotation pipeline (D.1)
	Audit / validation sub-phases
	Finalization & readiness
	Planned (not implemented)

High-Level Pipeline Overview



Version 1 is frozen. All active development is in Version2/.

Phase A - Framing Beam Extraction

Input: Framing plan DXF files (data/framing/)

Process: Extract beam marks, geometry, grid references from framing layers

Runner: docs/run_phases_abc.py

Output: phase_a/framing_beams.json, framing_validation.json

Phase B - Reinforcement Header Extraction

Input: Reinforcement DXF files (data/reinforcement/)

Process: Extract beam reinforcement header blocks and metadata

Runner: docs/run_phases_abc.py

Output: phase_b/reinforcement_headers.json

Phase C - Beam Cell Builder

Input: Framing beams + reinforcement headers

Process: Build beam cell grid; associate headers with framing beams

Runner: docs/run_phases_abc.py

Output: phase_c/beam_cells.json, beam_cells_validation.json

Debug: phase_c_debug/ - cell & sketch debug JSON + DXF overlays

Phase C.5 - Sketch Ownership

Input: Beam cells, sketches, header occurrences

Process: Assign sketch regions to beam occurrences

Runner: run_sketch_ownership.py

Output: phase_c5/sketch_ownership.json, header_occurrences.json

Phase D.1 - Raw Annotation Extraction

Input: Reinforcement DXF + sketch ownership
Process: Extract TEXT/MTEXT annotations inside sketch regions
Runner: run_phase_d1_annotations.py
Output: phase_d1/beam_annotations_raw.json

D.1 Sub-phases (ownership & classification)

- * **D.1.1 Ownership Audit**
Audit annotation-to-sketch ownership assignments
- * **D.1.2 Spatial Validation**
Validate annotation positions relative to sketch geometry
- * **D.1.3 Region Validation**
Confirm annotations fall within ownership regions
- * **D.1.3.1 Boundary Leakage**
Detect annotations leaking across region boundaries
- * **D.1.4 Reassignment**
Reassign mis-owned annotations to correct sketches
- * **D.1.5 Post-Reassignment Validation**
Validate ownership after reassignment
- * **D.1.6 Type Classification**
Classify: BAR, STIRRUP, ANCHORAGE, SFR, DIMENSION, NOTE

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Phase D.1.6A / D.1.6B - Coverage & Entity Survey

D.1.6A: Coverage audit - TEXT/MTEXT vs ownership (100% coverage confirmed)
D.1.6B: DXF entity survey - stirrups/Ld/dims live in DIMENSION entities
Key finding: Engineering callouts not fully captured by TEXT/MTEXT alone

Phase D.1.7 - DIMENSION Extraction

Input: Reassigned annotations + ownership + DIMENSION entities
Process: Extract DIMENSION overrides; integrate with ownership
Output: phase_d1_7/beam_annotations_extended.json (186 annotations)
Counts: BAR=97, STIRRUP=22, ANCHORAGE=6, DIMENSION=53, SFR=7

Phase D.1.7A - Dimension Source Audit

Purpose: Trace DIMENSION text source: override vs measurement
Finding: 687/688/530/537 = MEASUREMENT_VALUE (AutoCAD geometry)
Finding: Engineering text = DIMENSION_OVERRIDE (2L-Y8@100C/C, Ld, etc.)
Output: phase_d1_7a/dimension_source_audit.json

Phase D.1.7B - Engineering Annotation Filter

Input: Extended annotations + dimension source audit
KEEP: BAR, STIRRUP, ANCHORAGE, SIDE_FACE_REINF engineering text
REJECT: MEASUREMENT_VALUE (687, 688, 530, 537?)
GEOMETRY QA: 500, 1900, 2150 -> separate geometry channel
Output: phase_d1_7b/engineering_annotations.json (132 retained)

Phase D.1.7C - Integrity Audit (read-only)

Audits:	Fragments, stirrup completeness, anchorage, SFR, duplicates, types
Finding:	1 false rejection: 'd' fragment near Ld+10db on B18_S1
Finding:	29 duplicate ownership (shared template geometry)
Finding:	2 partial SFR notes (IGNORE_NOTE policy)
Result:	Parser readiness PASS (1.52% questionable)
Output:	phase_d1_7c/ - integrity reports + validation

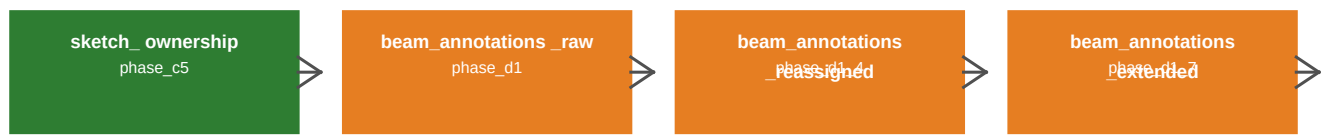
Phase D.1.7D - Dataset Finalization

Resolution:	'd' -> MERGED_WITH_ANCHORAGE (Ld+10db), IGNORE_FRAGMENT
Deduplication:	50 entries removed (same text+position across sketches)
SFR policy:	VALID_SFR=PARSE, PARTIAL_SFR=IGNORE_NOTE
Final counts:	80 parser-ready: BAR=49, STIRRUP=22, ANCHORAGE=6, SFR=3
Status:	READY_FOR_D2 - questionable annotations = 0
Output:	phase_d1_7d/engineering_annotations_final.json
Policies:	d2_parser_policy.json, sfr_parsing_policy.json

Phase D.2 - Planned Next Step

Input:	engineering_annotations_final.json + parser policies
Goal:	Parse BAR qty/dia, stirrup spacing, anchorage, SFR bar info
Status:	NOT STARTED - awaiting review approval

Key Data Flow - Annotation Pipeline



Validation Gates

- * **A/B/C**
Framing & cell validation JSON per phase
- * **D.1.2-D.1.5**
Spatial, region, boundary, post-reassignment checks
- * **D.1.7B**
PASS - no measurement leakage into engineering set
- * **D.1.7C**
WARN -> integrity issues documented
- * **D.1.7D**
PASS - READY_FOR_D2

Runner Command Reference

```
cd Version2  
$env:PYTHONPATH=""
```

Foundation

```
python docs/run_phases_abc.py  
python run_sketch_ownership.py
```

Annotation pipeline

```
python PY/run_phase_d1_annotations.py  
python PY/run_phase_d11_annotation_audit.py  
python PY/run_phase_d12_spatial_validation.py  
python PY/run_phase_d13_region_validation.py  
python PY/run_phase_d131_boundary_leakage.py  
python PY/run_phase_d14_reassignment.py  
python PY/run_phase_d15_post_reassignment_validation.py  
python PY/run_phase_d16_annotation_classification.py  
python PY/run_phase_d16a_annotation_coverage.py  
python PY/run_phase_d16b_entity_survey.py  
python PY/run_phase_d17_dimension_extraction.py  
python PY/run_phase_d17a_dimension_source_audit.py  
python run_phase_d17b_engineering_filter.py  
python run_phase_d17c_integrity_audit.py  
python run_phase_d17d_engineering_dataset_finalization.py
```